Jump process

Simulation of Lévy processes

Cathérine MONOUE KONGA

Master's student in Financial Mathematics at university of Luxembourg

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supervisor: prof. Mark PODOLSKIJ

Outline

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Introduction to Lévy processes
Motivation
Definition, structure and component
Simulation of Lévy process
Application in option Pricing
Up-and-out call option
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1. Introduction to Lévy processes
   - Why are we interested in Lévy process?
   - What is a Lévy process (definition,structure, and components)?
   - How are Lévy processes's characteristics used for simulation?
   - Simulation:Lévy process with Laplace jump size and λ intensity and drift b and brownian motion B_t
2. Application in option pricing : pricing of up-and-out call option
   - What is a up-and -out call option ?
   - Link between lévy processes properties and up-and-out call option price?

Why are we interested in Lévy processes?

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- Most real-world phenomena are continuous and sometimes exhibit sudden change.
- However, their observations remains discrete
- Researchers are looking for mathematical processes that can connect discrete observations to underlying continuous distribution dynamics and capture potential discontinuities.
- Lévy processes play a central role in this context, through their properties which make them particularly well suited for modelling continuous-time phenomena with discontinuities. We will see why as we explore their characteristics.

What is a Lévy process?

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Definition (Lévy process)

Let $(X_t)_{t\geq 0}$ be a stochastic process on $(\Omega, \mathcal{F}, \mathbb{P})$ with values in $\mathbb{R}^d$ such that $X_0 = 0$ and $(X_t)_{t>0}$ is right continuous with limit at left.
$(X_t)_{t>0}$ is called a **Lévy process**, if the it have the following properties:

- Independent increments: for every $t_0, ..., t_n$ increasing sequence of times, $X_{t_0}, X_{t_1} - X_{t_0}, ..., X_{t_n} - X_{t_{n-1}}$ are independents.
- stationary increments: The law of $X_{t+h} - X_t$ does not depend on t.
- Stochastic continuity: $\forall \epsilon > 0, \lim_{h\to 0} \mathbb{P}\big(|X_{t+h} - X_t| \geq \epsilon\big) = 0$

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property 1: infinite divisibility

A probability distribution F on $\mathbb{R}^d$ is **infinitely divisible** if for any integer $n \geq 2$ there exist n independent,identical distributed random variables $Y_1, ..., Y_n$ such $\sum_{i \geq 1} Y_i$ has the same distribution as F

proof: infinite divisibility

Let $t > 0$, $n > 2$, let's divide the $[0, t]$ in n equal intervals of time and denote by $h = \frac{t}{n}$ the length of those intervals.

$$X_t = \sum_{i \geq 0} \left(X_{(i+1)h} - X_{ih}\right) = \sum_{i \geq 0} Y_i$$

Since X_t has independent increments, (Y_i) are independent. Using the fact that the increments are also stationary, we have that all Y_i have the same distribution F_h.

Lévy Ito formula

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Let $(X_t)_{t\geq 0}$ be a Lévy process on $\mathbb{R}^d$, and ν its Lévy measure such that:

- ν is a Radon measure on $\mathbb{R}^d$ 0 and verifies $\int_{|x|\leq 1} |x|^2\nu(dx) < \infty$ and $\int_{|x|\geq 1} \nu(dx) < \infty$
- The jump measure of X, denoted by J_X is a poisson random measure on $[0,\infty[\times\mathbb{R}^d$ with intensity measure $\nu(dx)dt$
- there exist a vector γ and a d-dimensional Brownian motion $(B_t)_{t\geq 0}$

then $\mathbf{X_t = \gamma t + B_t + X_t^l + \lim_{\epsilon\to 0} \widetilde{X}_t^\epsilon}$

the term in the expression are independent, X_t^l *is a compound Poisson process and the convergence of* $\lim_{\epsilon\to 0} \widetilde{X}_t^\epsilon$ is almost sure.

In others terms: $\mathbf{X_t = \gamma t + B_t + X_t^l}$

Brownian process

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A stochastic process B_t is said to be a **standard brownian motion** if:

- B_t is a Gaussian process
- $\mathrm{B}_0 \equiv 0$
- $\mathrm{E}\left[\mathrm{B}_t\right] = 0$
- $\mathrm{E}\left[\mathrm{B}_t \mathrm{B}_s\right] = min(t, s)$
- $t \mapsto \mathrm{B}_t$ is continuous

From the definition above, we deduce $\mathrm{B}_t \sim \mathcal{N}(0, t)$

Brownian process

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In order to simulate a Brownian motion on the interval $[0, T]$, we exploit its infinite divisibility. Specifically,

- we can generate increments by drawing independent samples $\Delta \mathrm{B}_{t_i} \sim \mathcal{N}(0, \frac{T}{n})$. Here n is the number of time steps we wish to use in $[0, T]$ and $0 \leq i \leq n$
- Then, at time t_i ,the Brownian motion is given by the cumulative sum of the increments: $\mathrm{B}_{t_i} = \sum_{j=1}^{i} \Delta \mathrm{B}_{t_j}$

Compound Poisson process

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Definition (Compound Poisson Process)

A compound process with intensity $\lambda > 0$ and jump size distribution f is a stochastic process N_t defined as: $X_t = \sum_{i=0}^{N_t} Y_i$

Y_i are i.i.d. random variables with distribution f and represent jump sizes. Moreover, N_t is a Poisson process with intensity λ, independent of $(Y_i)_{i\geq 1}$.

Step for simulation

- simulate waiting time $\tau_i \sim \text{Exp}(\lambda)$ until $\sum \tau_i < T$
- Count the number N_T of τ_i obtained and generate N_T Y_i
- for $t \in [0, T]$, we take the first n_t τ_i that enable their sum reach t.
- $X_t = \lambda t + \mathrm{B_t} + \sum_{k=1}^{n_t} Y_i$

Illustration of simulated Lévy process

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Let $I = [0, 1]$ be the time's interval;
We want to simulate a Lévy process in I with

- dirft 0.2
- intensity $\lambda = 5$
- $\sigma = 0.5$
- Let us also assume that the jump size $Y_i \sim \text{Laplace}(0, 0.3)$
- subdivision $n = 1000$

I will play a video here

Compound Poisson process

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Proposition

If $(\tau_i)_{i\geq 1}$ are independent exponential random variables with parameter λ then for any $\lambda > 0$ the random variable : $\mathrm{N}_t = \inf\{n \geq 1 \sum_{i=0}^{n} \tau_i > t\}$ follows a Poisson distribution with parameter λt.

<u>Proof</u>

Let's denote $S_n = \sum_{i=0}^{n} \tau_i,.$

$$\begin{aligned} N_t = n &\equiv S_n \leq t < S_{n+1} \\ &\equiv \text{n arrival have occured by time t} \\ \text{thus, } \mathbb{P}(N_t = n) &= \mathbb{P}(S_n \leq t < S_{n+1}) \end{aligned}$$

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$$\begin{aligned}\{S_n \leq t\} &= \{S_n \leq t, S_{n+1} \leq t\} \cup \{S_n \leq t, S_{n+1} > t\} \\ &= \{S_{n+1} \leq t\} \cup \{S_n \leq t < S_{n+1}\} \\ \mathbb{P}(S_n \leq t) &= \mathbb{P}(S_{n+1} \leq t) + \mathbb{P}(S_n \leq t < S_{n+1})\end{aligned}$$

therefore,

$$\mathbb{P}(N_t = n) = \mathbb{P}(S_n \leq t) - \mathbb{P}(S_{n+1} \leq t)$$

S_n follows a gamma distribution with parameter (n, λ) so:

$$\begin{aligned}\mathbb{P}(N_t = n) &= \left(1 - \exp(-\lambda t) \sum_{k=0}^{n-1} \frac{(\lambda t)^k}{k!}\right) - \left(1 - \exp(-\lambda t) \sum_{k=0}^{n} \frac{(\lambda t)^k}{k!}\right) \\ &= \exp(-\lambda t) \frac{(\lambda t)^n}{n!}\end{aligned}$$

Thus N_t follows a Poisson distribution with parameter λt

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Proposition

Let $(\tau_i)_{(i=1,\dots,n)}$ be independent exponential random variables with parameter λ and $S_n = \sum_{i=0}^{n} \tau_i$ then,

The law of $(S_1, \dots, S_n)$ knowing S_{n+1} has the same law as $(Y_1, \dots, Y_n)$ where Y_i is the order statistic of n uniformly distributed random variables.

Proof

$$\begin{cases} S_1 = \tau_1 \\ S_2 = \tau_1 + \tau_2 \\ \vdots \\ S_n = \tau_1 + \cdots + \tau_n \end{cases} \quad \text{with} \quad J = \begin{vmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 0 \\ 1 & 1 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & 1 \end{vmatrix} = 1$$

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$$F_{(S_1,\dots,S_n)}(s_1,\dots,s_n) = \int_{-\infty}^{t_1}\cdots\int_{-\infty}^{t_n} \lambda^n \exp\left(-\lambda\sum_{i=1}^{n}\tau_i\right)\, d\tau_1\cdots d\tau_n$$
$$= \int_{-\infty}^{t_1}\cdots\int_{-\infty}^{t_n} \lambda^n \exp(-\lambda s_n)\, d\tau_1\cdots d\tau_n$$

$$\lambda^n e^{-\lambda s_n} 1_{\{0<s_1<\cdots<s_n\}} = \lambda e^{-\lambda s_n}\frac{(-\lambda s_n)^{(n-1)}}{(n-1)!}\, 1_{\{s_n\geq 0\}}$$
$$\times \frac{(n-1)!}{s_n^{n-1}}\, 1_{\{0<s_1<\cdots<s_n\}}.$$

This property will be used later to simulate $(S_1,\cdots,S_n)$ by sampling n observations u_i from a uniform random variable on

Illustration of simulated Lévy process using uniform distribution

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Simulation step

- Sample a single observation from a Poisson distribution with parameter λt. Let N be the value sampled;
- Sample N obervations from uniform on $[0, T]$ and ordered them;
- simulate the jump size

$$\mathrm{X}_t = \lambda t + \mathrm{B_t} + \sum_{k=1}^{N} 1_{U_i < t} Y_i$$

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What is a up-and out call option

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Definition (Option)

An option is a contract which gives the right, not the obligation to its owner to buy or sell an asset at a fixed price at a chosen date.

Definition (Strike K)

A strike is the price at which the owner of a call(or put) buy(or sell) underlying asset when the option is exercised.

Definition (barrier B)

A barrier is a price level at which the option is activated or extinguished.

up-and-out call option

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Definition (up-and-out call option (strike K, barrier B))

It is an option which gives the right to buy an underlying asset at a strike K, if the price of the underlying asset does not match or exceed a chosen level B over the contract's lifetime.

Definition (Payoff of up-and-out call option)

Let K be the strike, B the barrier, T the maturity and P_{up-out} the payoff of the up-and-out call option. Let S_t denote the price of the asset at time t.

$$P_{up-out} = (S_T - K)_+ 1_{\max\limits_{0 \leq t \leq T} S_t < B}$$

Levy exponential model

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Definition

The Lévy exponential model is a framework in which the asset price is given by :

$$S_t = S_0 \exp(X_t) \quad \text{with } X_t \text{ being a Lévy process}$$

Definition (Price of a call option)

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ be an arbitrage-free market with constant interest rate r. The filtration $\mathcal{F}_t$ represents the history of the asset price S_t, and $\mathbb{P}$ denotes the risk-neutral probability measure. The price of the call option with payoff C and maturity T is given as:

$$\Pi_t(C) = \exp(-r(T-t))\mathrm{E}^{\mathbb{P}}[C|\mathcal{F}_t] \quad t \in [0, T]$$

Lévy exponential mode

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Pricing the option means that we are looking for the price of the option at $t = 0$

$$\Pi_0(C) = \exp(-r(T))\mathrm{E}^{\mathbb{P}}\left[(S_T - K)_+ 1_{\max_{0\le t\le T} S_t < B}\right]$$

because there is no information available at the beginning. Our next step is to find a way to estimate $1_{\max_{0\le t\le T} S_t < B}$

- splits $[0, T]$ at jump times;
- uses that, between jumps, a Lévy process coincides with a Brownian motion and hence satisfies the Markov property;
- and applies a known result for the joint distribution of a Brownian motion and its supremum.

Using these facts lead us to:

$$\Pi_0(C) = \exp(-r(T))\mathrm{E}^{\mathbb{P}}\left[(S_T - K)_+ \prod_{i=1}^{N} 1_{\mathrm{X}(\tau_i)<b}\left(1 - \exp\left(\frac{2(\mathrm{X}_{\tau_i-} - b)(\mathrm{X}_{\tau_{i-1}} - b)}{(\tau_i - \tau_{i-1})\sigma^2}\right)\right)\right]$$

Simulation of an up-and-out option under an exponential Lévy model

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- simulate the jump times and determine how many jumps occur between 0 and each jump time;
- simulate the Brownian component at each jump time;
- compute the value of the function appearing inside the expectation;
- iterate the entire simulation and take the empirical mean of these function values

Results

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Let $I = [0, 1]$ be the time's interval; We want to simulate the price of the up-and out call option using a the exponential Lévy process in I.

- drift $b = 0.05$, $\sigma = 0.2$
- $\lambda = 5$
- jump size $Y_i \sim \text{Laplace}(0, 0.1)$
- $S_0 = 100$, $K = 50, H = 80$
- Monté Carlo path $= 500$
- interest rate $r = 5\%$

We get $C = 58.23$

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Thank You.

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